Code No: 118BA **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, July - 2019
DATABASE SECURITY
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART - A

(25 Marks)

1.a)	What is man in the middle attack?	[2]
b)	Give a short note on transaction security.	[3]
c)	Write Tranquility principle.	[2]
d)	What is dial-p gate and root gate.	[3]
e)	What is design pattern and write its usage	[2]
f)	What is unencrypted index? Write its purpose.	[3]
g)	What is complex object?	[2]
h)	Write key differences between frame based and object oriented databases.	[3]
i)	Define weak authorization base.	[2]
j)	What is active database? Write its features.	[3]

PART - B

(50 Marks)

2.a)	What is access matrix model? Explain its major parts and implementation.	
b)	Explain Take grant model in detail.	[5+5]
	OR	
3.a)	Explain in detail about database inference.	
b)	Give a note on database auditing.	[5+5]
4.a)	What is Biba model? Write its features.	
b)	Explain the working principle of Biba model for database security.	[5+5]
	OR	
5.a)	Explain memory protection mechanisms.	
b)	Explain, how database security is provided using Dion's model.	[5+5]
6.a)	Explain design considerations for security software.	
b)	Explain different categories of inference.	[5+5]
	OR	
7.a)	Describe different types of attacks for inference control evaluation.	
b)	Write a note on ASES system.	[5+5]

JNTUH USED 19-07-2019 AM

8.a) Explain security model for frame based model.
b) Write a note on the core concepts of object orient database. [5+5]

OR

9.a) Write the security issues in object oriented database.
b) Explain SORION model for protecting object oriented database. [5+5]

10.a) Explain ORION model architecture with a neat sketch.
b) Explain model for protection of active databases. [5+5]

OR

11.a) Write a note on Kogan's Model.
b) Explain about authorization in detail and its types. [5+5]

---ooOoo---

JNTUH USED 19-07-2019AM